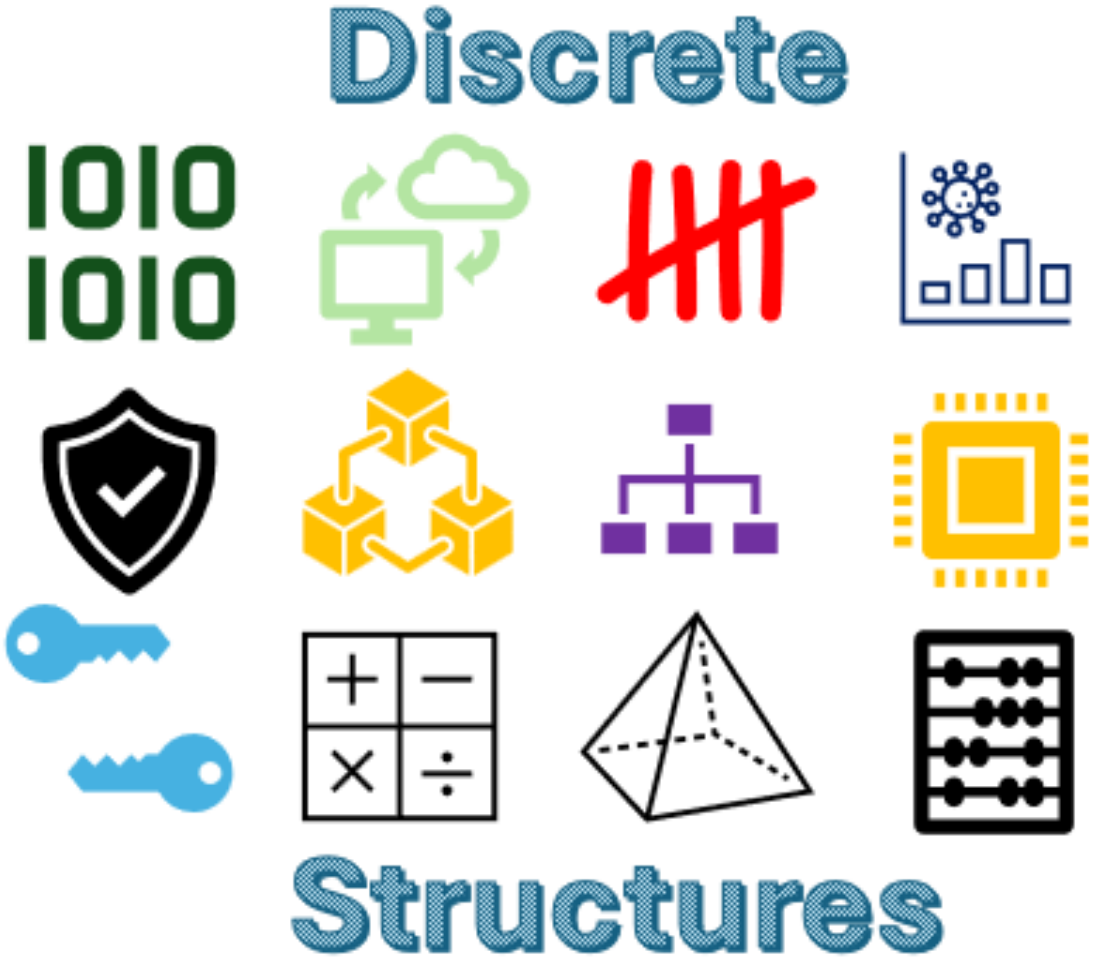
COMP 2310: Discrete Structures
Fall 2026

Chapter 3: Counting



Combinations

- Combination vs permutations
  - Combinations: order does not matter
  - Permutations: order matters
  - A = {1, 2, 3} is same as A = {1, 3, 2} in Combinations
  - A = {1, 2, 3} are not same as A = {1, 3, 2} in Permutations
- Number of permutations is always larger than the number of combinations
  - Why?
  - If order matters: you need to consider all regardless of order (permutations)
  - If order does not matter: you will discard the sets having the same elements but in different order (combination).

Combinations

- Let *A* be any set with *n* elements and $0 \le r \le n$. How many different subsets of *A* are there, each with *r* elements?
  - **In other words:** combination of *A*, taken *r* at a time.
    - Assume n = 4, therefore, $0 \le r \le 4$ (r can take have 0 to 4 elements)
    - A = {0, 1, 2, 3}, or A = {2, 3, 5, 1}
    - As order does not matter, A = {0, 1, 2, 3}, and A = {2, 3, 0, 1} are the same
  - **Going to back to the last lecture in Permutations: In other words:** how many ways you can pick 2 balls (r = 2) at a time from 3 balls (n = 3) [**red, green, blue** balls]
  - How did we calculate?
    - ${}_nP_r = {}_3P_2 = \frac{n!}{(n-r)!} = \frac{3!}{(3-2)!} = 6$ (This is for permutation where order matters)
    - ${}_nC_r = \frac{n!}{r!(n-r)!} = \frac{3!}{2!(3-2)!} = 3$ (This is for combination where order does not matter)

Combinations or permutations?

- Examples:
  - A. Compute the number of distinct (no repeat) five-card that can be dealt from a deck of 52 cards.
  - B. Compute the number of distinct (no repeat) sequences (order matters) of five cards that can be formed from a deck of 52 cards.

  - For both the problems: n = 52, r = 5
  - A. **Combinations (order does not matter):** use formula = ${}_{n}C_{r} = {}_{52}C_{r} = \frac{n!}{r!(n-r)!}$
  - B. **Permutations (order matters):** use formula = ${}_{n}P_{r} = {}_{52}P_{r} = \frac{n!}{(n-r)!}$.

Combinations

- Suppose *k* selections are to be made from *n* items without regard to order and repeats are allowed, assuming at least *k* copies of each of the *n* items. The number of ways these selections can be made is ${}_{(n+k-1)}C_k$.
- Example: 3 balls are to be made from 5 balls without regard to order (combinations) and repeats are allowed (n+k-1)
  - ${}_{5+3-1}C_3 = {}_7C_3$
- *k+n-1* rather than *k* why?
- Stars and bars problem.
  - How many bars do you need to separate 5 balls?
    - 4 bars (ball1 | ball 2 | ball 3 | ball 4 | ball 5) [n-1 bars]
    - How many balls to be selected?
      - 3 balls (which is k), can be represented by *.
  - Therefore, you have n-1 bars and k starts ➔ n – 1 + l

Example

- In how many ways can a prize winner choose three CDs from the Top Ten list if repeats are allowed?

k = 3, n = 10, repeats allowed (n – 1 + k), order does not matter (combination)

$$_{n-1+k}C_3 = {}_{12}C_3 = \frac{12!}{3!(12-3)!} = 220$$

Altogether: truth table

Scenario	Order matters	Repetition allowed	Type	Formula
Choose r items from n (order matters, no repeats)	Yes	No	Permutation	$\frac{n!}{(n-r)!}$
Choose r items from n (order matters, repeats)	Yes	Yes	Permutation	n^r
Choose r items from n (order doesn't matter, no repeats)	No	No	Combination	$\frac{n!}{r!\,(n-r)!}$
Choose r items from n (order doesn't matter, repeats)	No	Yes	Combination	$\frac{(n-1+r)!}{r!\,(n-r)!}$

Altogether: clear your confusions

- 1. How many 5-digit passwords can be generated from 0-9 digits?
- 2. How many 5-digit distinct passwords can be generated from 0-9 digits?
- 3. How many ways can you choose 5 digits at a time from 0-9 to generate a password where order matters and repetition is not allowed?
- 4. How many ways can you choose 5 digits at a time from 0-9 to generate a password where order matters and repetition is allowed?
- 5. How many ways can you choose 5 digits at a time from 0-9 to generate a password where order does not matter and repetition is not allowed?
- 6. How many ways can you choose 5 digits at a time from 0-9 to generate a password where order matters and repetition is not allowed?
- 7. How many ways can you choose 5 digits at a time from 0-9 to generate a password where order does not matter and repetition is allowed?
- 8. How many total ways can you log in if you can use any of your 2 email addresses along with any of your five phone number?

Example

- 1. How many 5-digit passwords can be generated from 0-9 digits?

  r = 5

  N = 10

  N^r = 10 X 10 X 10 X 10 X 10 =?

Example

- 2. How many 5-digit distinct passwords can be generated from 0-9 digits?
  - Distinct: each item in the set is unique

    10 X 9 X 8 X 7 X 6 =?
  - Same as permutation: ${}_{10}P_5$

Example

- 3. How many ways can you choose 5 digits at a time from 0-9 to generate a password where order matters and repetition is not allowed?
  - Same as 2, apply the same permutation formula, or multiply

Example

- 4. How many ways can you choose 5 digits at a time from 0-9 to generate a password where order matters and repetition is allowed?
  - n^r

Example

- 5. How many ways can you choose 5 digits at a time from 0-9 to generate a password where order does not matter and repetition is not allowed?
  - Order does not matter: combinations
  - See third row in the table

Example

- 6. How many ways can you choose 5 digits at a time from 0-9 to generate a password where order matters and repetition is not allowed?
  - See the first row in the table

Example

- 7. How many ways can you choose 5 digits at a time from 0-9 to generate a password where order does not matter and repetition is allowed?
  - Order does not matter: combination
  - See the fourth row in the table

Example

- 8. How many total ways can you log in if you can use any of your 2 email addresses along with any of your five phone number?
- N x N ways = 2 x 5 = 10